

STAI-X Award B -- Run Report

run_id: 20260614_072322

Task type: Regression

Target: rate_per_10000_ed_visits

Primary metric: block_mae (scoring restricted to 3 overdose categories)

TL;DR: GBDT specialist selected, OOF block_mae = **1.6509** (24.5% over the category-mean baseline); primary caveat is strong residual heteroscedasticity (corr = 0.617) that signals systematic underprediction of high-rate observations.

1. Executive Summary

The training dataset is data/train/dose_sys_train.csv with **31,416 rows** and **4 columns**, covering 77 monthly periods (2019-01 through 2025-05) across 51 jurisdictions and 8 overdose categories. The task is regression: predict rate_per_10000_ed_visits (suspected nonfatal overdose ED visits per 10,000 total ED visits) for 918 rows in the validation frame. The evaluation metric is MAE restricted to three scored categories (all_drugs, all_opioids, all_stimulants).

The selected model is the GBDT specialist stack (stack(tuned(catboost)+tuned(hist_gradient_boosting)+catboost_log)), which achieves a **block_mae of 1.6509** on the canonical forward-expanding 5-fold OOF restricted to the 3 scored categories -- a **24.5% improvement** over the category-mean baseline (2.1855) and a **2.7% improvement** over the deterministic floor (1.6970). Robust generalization selection was not used; classic keep-best selection on common OOF was applied via NNLS blending. The primary data limitation is heavy right-skew of the target (skewness = 2.70) combined with high missing rates in the prediction frame for unemployment_rate (54.6%) and labor_force (58.0%).

2. Task Interpretation

Task type is **regression**, determined from DATA_DESCRIPTION.md (primary authority; confidence = 0.97). The target column is rate_per_10000_ed_visits. The row identifier is row_id. The evaluation metric is MAE, implemented as block_mae with block column overdose_category. Output format is continuous. The required submission has exactly two columns (row_id, rate_per_10000_ed_visits) matching the row order of data/sample_submission.csv.

The data has a panel structure: each row is uniquely identified by (period_id, jurisdiction, overdose_category). Prediction rows cover 6 validation period_ids that are strictly later than all training periods, with no temporal overlap.

3. Data Overview

Training file: data/train/dose_sys_train.csv -- **31,416 rows, 4 columns.**

Column	Dtype	Missing rate (train)	Notes
period_id	object	0.0%	Opaque chronological identifier; 77 unique values
jurisdiction	object	0.0%	51 unique US jurisdictions
overdose_category	object	0.0%	8 categories; 3 are scored
rate_per_10000_ed_visits	float64	0.0%	Target; range [0.05, 79.99], mean 6.35, std 7.97

Covariates are supplied in a separate file (data/train/covariates.csv, 3,927 rows, 12 columns) joined on (period_id, jurisdiction).

Column	Dtype	Missing rate (train)	Notes
period_id	object	0.0%	Join key
jurisdiction	object	0.0%	Join key
unemployment_rate	float64	0.0%	Macro indicator
labor_force	int64	0.0%	Size indicator
temp_avg_f	float64	~5.9%	Imputed from periodxjurisdiction median
precip_in	float64	~5.9%	Imputed from periodxjurisdiction median
gtrends_overdose	float64	0.0%	Highest-influence covariate
gtrends_fentanyl	float64	0.0%	Distribution shift flagged
gtrends_naloxone	float64	0.0%	Distribution shift flagged
gtrends_opioid	float64	0.0%	Distribution shift flagged
gtrends_methamphetamine	float64	0.0%	
state_doh_release	object	~29.6%	High-cardinality text; TF-IDF SVD applied

Prediction file: data/val/covariates.csv -- **357 rows, 12 columns** (same schema as train covariates). The sample submission has 918 rows and 5 columns; its extra context columns (period_id, jurisdiction, overdose_category) are not part of the required output. Key schema differences: unemployment_rate is 54.6% missing and labor_force is 58.0% missing in the prediction frame; state_doh_release is 32.2% missing. Image sidecars (data/val/images/mat_density/, 357 PNGs) are present and were used.

4. Preprocessing and Feature Engineering

Excluded columns: rate_per_10000_ed_visits (prediction target) and row_id (row identifier) were excluded from the feature matrix. period_id and jurisdiction are used as keys and grouping variables, not as raw string features. overdose_category is used only as a one-hot-encoded feature and grouping key, never as a raw string in model input.

Imputation: temp_avg_f and precip_in are imputed from periodxjurisdiction median. unemployment_rate and labor_force are imputed from jurisdiction median for prediction-frame rows (where missing rates exceed 54%). state_doh_release missing values receive a zero-vector SVD embedding.

Encoding: overdose_category is one-hot encoded (8 binary columns). jurisdiction is target-encoded per fold. state_doh_release is vectorized via TF-IDF followed by 20-component SVD, fitted per fold on training rows.

Datetime-like columns: period_id is an opaque hash; its chronological order was resolved from validation_strategy.json.holdout_parameters.period_order. The derived feature period_id__ordinal_rank (integer 0-76) was added and validated via the feature audit. No raw period_id string was used as a model feature.

Data-pattern findings: The panel is a time series with 77 periods and 408 unique (jurisdiction, overdose_category) series, each of full length 77. Strong target autocorrelation -- lag-1 $r = 0.562$, lag-3 $r = 0.521$, lag-6 $r = 0.465$, lag-12 $r = 0.422$ -- motivated lag features at horizons 1, 3, 6, and 12, plus rolling means and standard deviations. Among covariates, gtrends_overdose ($|r| = 0.258$), gtrends_methamphetamine ($|r| = 0.226$), and gtrends_opioid ($|r| = 0.185$) are among the strongest predictors of the target, and were prioritized for interaction terms with overdose_category. These correlations guided the feature plan; they are associations, not causal claims.

All transformations were fitted on the training split only.

Image features: 17 scalar summaries were extracted from mat_density PNG sidecars (viridis colormap, inverted to scalar field) using numpy/PIL/matplotlib -- pixel statistics (img__mean, img__std, img__p10, img__p50, img__p90, img__high_frac, img__cover, img__cmass_x, img__cmass_y) and 8 SVD components (img__svd0-img__svd7). These are static per (jurisdiction, period_id) observations not derived from the target.

The feature set totals **102 columns** across 4 groups: direct numeric + OHE + interactions (54), per-fold target aggregates + text SVD (30), image features (17), and datetime-derived (1). No groups were pruned by the ablation gate.

5. Validation Strategy

Chosen strategy: forward_expanding_time, selected by the validation-and-schema-guardian based on the panel's strict temporal structure -- validation periods are all strictly after training periods with no overlap. An expanding-window backtest (train on past periods, validate on future horizon blocks of size 6) directly simulates the train → hidden-evaluation gap. GroupKFold over random periods would allow future periods to inform predictions of the past (forward-peeking leakage), producing optimistic OOF estimates.

Parameters: 5 folds, horizon = 6, time column = period_id, group column = jurisdiction. The final fold trains on the first 71 periods and validates on the 6 most recent training periods, providing the closest analogue to the hidden test gap.

Scoring subset: {run_id}_cv_folds.json.scoring_restricted = true. OOF block_mae is computed over the 3 scored categories only (all_drugs, all_opioids, all_stimulants), which aligns the CV metric directly with the leaderboard. Train-only categories (benzodiazepine, cocaine,

fentanyl, heroin, methamphetamine) participate in model training and provide lag-feature context but do not contribute to the reported OOF metric.

Leakage risks logged: Medium -- per-fold target aggregates for fold 0 (no prior training rows) fall back to global mean, which is a mild forward-data contamination for fold-0 rows only. Medium -- TF-IDF SVD coordinates differ across folds (no shared embedding space). Low -- imputation medians for numeric columns computed on full training set before fold split. Low -- distribution shift in `gtrends_fentanyl`, `gtrends_naloxone`, `gtrends_opioid` between train and prediction frame. None of these rise to blocking severity.

Validation limitations: OOF covers only the last ~30 training periods (5 folds x horizon 6); earlier rows do not contribute OOF signal. Distribution shift in some covariate columns means OOF MAE may slightly underestimate leaderboard MAE.

6. Candidate Models

Three specialist families were evaluated in parallel on the canonical shared folds, then combined via NNLS blending. The GBDT stack won outright; the blend offered no improvement over the best individual family.

Model	OOF block_mae (restricted)	Blend weight (NNLS)	Selected model architecture	Notes
Baseline (category mean)	2.1855	--	--	Reference baseline
Floor (pipeline)	1.6970	0.138	<code>stack(tuned(catboost)+catboost_log+lightgbm_log)</code>	Seeded by Step-6A main.py
GBDT specialist	1.6509	0.455	<code>stack(tuned(catboost)+tuned(hist_gradient_boosting)+catboost_log)</code>	Selected; monotonic constraints applied
Trees specialist	1.7032	0.000	<code>stack(tuned(random_forest)+extra_trees_log+random_forest)</code>	Dead weight; NNLS weight = 0
Linear specialist	1.6946	0.407	<code>stack(elastic_net+ridge_log+ridge)</code>	cv_std = 0.063, relative_stability = 6.7%
NNLS blend	1.6512	--	GBDT + linear + floor	Negligibly worse than GBDT alone

The NNLS blend (block_mae = 1.6512) is marginally worse than the GBDT alone (1.6509), so the ensemble-meta agent correctly selected GBDT as the promoted candidate.

Note: the specialist agents' self-reported `cv_score` fields (e.g., `gbdt` = 0.8467) reflect an internal scoring scope that differs from the canonical 3-category restricted OOF; only the `ensemble_meta` canonical scores are comparable across candidates.

7. Selected Model and Predictions

The GBDT specialist (`stack(tuned(catboost)+tuned(hist_gradient_boosting)+catboost_log)`) was promoted based on strictly outperforming the floor on the same 4,590 common OOF rows (`block_mae` = 1.6509 vs floor = 1.6970), restricted to the 3 scored overdose categories. The improvement over the floor is 2.7%; the improvement over the category-mean baseline is 24.5%. No ensemble blend improved on the GBDT specialist alone.

Result: OOF `block_mae` = **1.6509** (24.5% improvement over category-mean baseline of 2.1855; 2.7% improvement over deterministic floor of 1.6970).

The floor was not retained; the GBDT specialist beat it on common OOF and was promoted to `submission.csv`. The NNLS blend was computed for completeness but its score (1.6512) was marginally worse, confirming single-candidate promotion was correct.

Submission validation (`submission_validation.json`): `columns_ok` = true, `row_count_ok` = true, `row_id_alignment_ok` = true, `all_finite` = true, overall verdict = **PASS**.

8. Overfitting and Generalization Controls

8.1 Validation strategy rationale

The forward-expanding time CV was selected because the panel has 77 training periods strictly earlier than the 6 validation periods (no temporal overlap). Random or group-only splits would allow future data to inform past predictions. The five-fold expanding window accumulates all available history per fold, matching production inference.

8.2 Train vs. validation gap

No train score was reported in any specialist artifact (`agent_gbdt.json`, `agent_trees.json`, `agent_linear.json` all report only `cv_score`). A direct train-val gap cannot be computed. The generalization-reviewer classified overfit risk as **moderate**, citing heteroscedasticity correlation = 0.617 (Q4 mean residual = -0.97 vs Q1 = +0.23), indicating the model systematically underpredicts high-rate series. The robust adjusted score was not applied; classic keep-best selection was used.

8.3 Number of validation splits

5 splits (`validation_strategy.json.holdout_parameters.n_splits` = 5), with horizon = 6 periods per split.

8.4 Model complexity summary

All three specialist families use stacked ensembles of 2-3 sub-models with early stopping or regularization. The winning GBDT stack combines tuned CatBoost, tuned HistGradientBoosting, and CatBoost with log-target transform. The linear specialist (elastic net + ridge) provides a lower-complexity reference and received 40.7% NNLS weight despite inferior individual OOF, suggesting partial

complementarity in OOF residuals. The trees family (random forest + extra trees) received zero NNLS weight and is identified as dead weight.

8.5 Leakage audit result

overfitting_leakage_audit.json verdict: **WARN** (no HIGH-severity leakage). CV strategy is correctly implemented as forward-expanding. Two medium findings: (1) fold-0 target aggregate imputation uses full-train global mean rather than period-restricted prior data; (2) numeric imputation medians computed on full training set before fold split. One low finding: hardcoded column names in interaction construction within feature_pipeline.py. None of these block delivery.

8.6 Prediction sanity result

prediction_sanity.json verdict: **PASS** (all 7 checks passed). 918/918 predictions finite; 0 negative values; distribution KS vs scored training target = 0.086; mean ratio pred/scored_train = 0.975 (2.5% off); no clipping artifacts; row_id order matches sample submission.

8.7 Final model selection rationale

From analysis_review_1.json: "No blocking HIGH-severity leakage. Best CV = 1.6509 (24.5% improvement over baseline = 2.1855). CLAUDE.md Step 6E mandates next_action = stop_and_report unconditionally in single-round mode. Shipping current best submission." approved_for_final = true.

8.8 Repair rerun status

supervisor_gatekeeper.json was not produced in this run. No repair rerun was triggered.

9. Hardcoding Audit

Pre-run audit (hardcoding_audit_pre_20260614_072322.json): conducted over src/, scripts/, and repository source files before authored code was written.

Post-run audit (hardcoding_audit_post_20260614_072322.json): scanned src/, scripts/, and outputs/scratch/20260614_072322/feature_pipeline.py. Verdict: **fail**.

Classification	Count
Unacceptable	22
Risky	58
Acceptable	15

The 22 unacceptable findings are concentrated in outputs/scratch/20260614_072322/feature_pipeline.py, where column names such as gtrends_overdose, overdose_category, and jurisdiction appear as literal string references in interaction and lag construction logic rather than being resolved dynamically from analysis_plan.json

or `spec_parse.json`. The 58 risky findings include domain vocabulary terms (overdose, opioid) appearing in comments and variable names. These do not affect the submission for this run since the columns match the actual data, but they represent policy violations that should be resolved in a future iteration.

10. Limitations

Missing data: `unemployment_rate` (54.6% missing in prediction frame) and `labor_force` (58.0% missing) are imputed from jurisdiction median. `state_doh_release` (32.2% missing in prediction frame) receives zero-vector SVD embedding. These high missing rates introduce imputation uncertainty in the prediction frame that is not reflected in OOF scores, which are computed on training data where missingness is low.

Datetime features: `period_id` is an opaque hash; its chronological order was resolved from a manually validated period map. If the ordering is incorrect for any period, temporal features and lag features will be wrong for affected rows. No sub-daily features were applicable (monthly data).

Model limitations: The GBDT specialist shows strong heteroscedasticity ($\text{corr} = 0.617$): high-rate series are systematically underpredicted (Q4 mean residual = -0.97). Applying $\log_{1p}$ transform uniformly to all families (rather than only to select sub-models) is the highest-leverage fix identified by reviewers.

Validation limitations: OOF is computed over the last 30 training periods only (5 folds x 6 horizon); early-period rows contribute no OOF signal. Distribution shift in `gtrends_naloxone` (train mean = 47 vs prediction-frame mean = 64) may cause the OOF MAE to underestimate leaderboard MAE.

Overfitting risk: Moderate. No train score was reported by specialists, so the train-val gap cannot be directly quantified. The generalization reviewer notes a 1.95x discrepancy between the GBDT specialist's self-reported `cv_score` (0.8467, computed over all 8 categories) and the canonical meta OOF (1.6509, restricted to 3 scored categories); the canonical figure is the reliable benchmark.

Trees family: The trees specialist received zero NNLS blend weight (OOF = 1.7032 vs floor = 1.6970), offering no value over the floor.

Hardcoding policy violations: 22 unacceptable findings in the authored feature pipeline code (literal column names). These do not affect this run's predictions but must be addressed before generalizing the pipeline to new datasets.

Generalizability: Generalization to future data has not been validated. All findings describe statistical associations. No causal claims are made.

11. Appendix

Feature list (102 columns from `feature\_spec.json`)

Direct numeric (9): `unemployment_rate`, `labor_force`, `temp_avg_f`, `precip_in`, `gtrends_overdose`, `gtrends_fentanyl`, `gtrends_naloxone`, `gtrends_opioid`, `gtrends_methamphetamine`

One-hot encoded (8): overdose_category__all_drugs, overdose_category__all_opioids, overdose_category__all_stimulants, overdose_category__benzodiazepine, overdose_category__cocaine, overdose_category__fentanyl, overdose_category__heroin, overdose_category__methamphetamine

Datetime-derived (1): period_id__ordinal_rank

Per-fold target aggregates (10): jurisdiction__target_enc, tgt_mean/std/median__jurisdiction_overdose_category (3), tgt_mean/std/median__jurisdiction (3), tgt_mean/std/median__overdose_category (3)

Target lags and rolling (9): rate_per_10000_ed_visits__lag1/3/6/12 (4), rate_per_10000_ed_visits__rollmean3/6/12 (3), rate_per_10000_ed_visits__rollstd6/12 (2)

Covariate lags/rolling (3): gtrends_overdose__lag1, gtrends_methamphetamine__lag1, gtrends_overdose__rollmean3

Interaction features (25): gtrends_overdose_x_overdose_category__* (8), gtrends_methamphetamine_x_overdose_category__* (8), gtrends_fentanyl_x_overdose_category__* (8), gtrends_overdose_x_period_rank (1)

Text SVD (20): state_doh_release__svd0 through state_doh_release__svd19

Image features (17): img__mean, img__std, img__p10, img__p50, img__p90, img__high_frac, img__cover, img__cmass_x, img__cmass_y, img__svd0-img__svd7

Excluded columns

From analysis_plan.json: rate_per_10000_ed_visits (target), row_id (row identifier), period_id and jurisdiction (used as keys/group columns only, not raw features).

Run metadata

Field	Value
run_id	20260614_072322
task_type	regression
metric	block_mae
random_state	42
modeling_mode	specialist
n_folds	5
horizon	6
target_transform	log1p

Artifact inventory

Artifact	Path
Feature parquet (train)	outputs/logs/20260614_072322_features_train.parquet
Feature parquet (pred)	outputs/logs/20260614_072322_features_pred.parquet
CV fold assignments	outputs/logs/20260614_072322_cv_folds.json
OOF predictions (GBDT)	outputs/logs/20260614_072322_oof_gbdtd.csv
OOF predictions (floor)	outputs/logs/20260614_072322_oof_floor.csv
Submission (final)	submission.csv
Feature pipeline	outputs/scratch/20260614_072322/feature_pipeline.py
Report Markdown	outputs/reports/20260614_072322_report.md
Report PDF	outputs/reports/20260614_072322_report.pdf